CROPSAGE PLANNING REFERENCE

Final MVP Crop Catalog Fields & Location Comparisons

A practical schema for comparing sourced crop requirements with location-specific temperature, soil, water, region, and planting evidence across Texas.

How the evidence is divided

Dimension	Stored in crop catalog	Compared with
Temperature	Crop thermal ranges and thresholds	FortyGuard recent/local heat plus NASA long-term climatology
Soil	Crop texture, pH, and drainage requirements	USDA SSURGO map-unit properties, with optional soil-test override
Water	Demand, drought tolerance, and irrigation dependency	NASA precipitation, SSURGO water storage, and farmer irrigation status
Region and season	Texas applicability, planting window, maturity	Farm coordinates, region mapping, and proposed planting date
Evidence quality	Sources, confidence, review date, assumptions	Used by validation and confidence; not compared as agronomic conditions

Scope note: SSURGO estimates what the soil can store and how it drains. It does not report current soil moisture, rainfall, irrigation access, or legal water availability.

Comparison model

CropSage compares stable catalog requirements with normalized location evidence. The deterministic scorer produces separate suitability, risk, and confidence outputs; missing evidence lowers confidence rather than being treated as favorable.

Complete Catalog Field Matrix

All required crop-catalog fields are listed in one continuous table. The header repeats when the table crosses a page boundary.

Group	Crop catalog field	Location evidence	Comparison or purpose
Identity	`crop_id`	None	Stable internal identifier used across scoring, citations, and UI state.
Identity	`common_name`	None	Farmer-facing display name.
Identity	`scientific_name`	None	Prevents ambiguity between crops with similar common names.
Identity	`production_use`	Farmer production objective (not API)	Separates production profiles such as grain corn and silage corn.
Identity	`variety_scope`	None	States whether requirements apply to a species, commercial type, or named cultivar.
Region	`supported_texas_regions`	Farm coordinate + Texas region resolver (pending); Open-Meteo geocoding confirms Plainview	Determines whether the crop profile is supported at the selected location.
Region	`regional_suitability`	Manual Texas Plains context; automated region resolver pending	Returns core, conditional, marginal, or unsupported.
Season	`planting_windows_by_region`	Resolved region + farmer proposed planting date (pending)	Checks whether the scenario date is inside a recommended planting window.

Group	Crop catalog field	Location evidence	Comparison or purpose
Season	`days_to_maturity`	Planting date + frost-free season length (pending)	Checks whether the crop can mature before damaging cold or seasonal limits.
Temperature	`optimal_temperature_range`	FortyGuard local heat + Open-Meteo recent/forecast + NASA climatology (Plainview rerun pending)	Measures whether expected baseline and recent local temperatures are favorable.
Temperature	`heat_stress_threshold`	FortyGuard threshold exceedance hours for the catalog threshold	Detects crop-specific threshold exceedance.
Temperature	`heat_exposure_duration`	FortyGuard exceedance hours, normalized exposure share, and persistence hours	Separates a brief spike from persistent heat exposure.
Temperature	`heat_sensitive_stages`	Planting date + derived crop stage (pending) + FortyGuard heat window	Applies stronger heat penalties during sensitive growth stages.
Temperature	`frost_sensitivity`	NASA daily minimum temperature after Plainview rerun; frost-free derivation pending	Identifies cold or frost risk for the proposed crop calendar.
Temperature	`temperature_unit`	FortyGuard, NASA, and Open-Meteo units normalized to Celsius	Ensures all values are normalized before comparison.
Temperature	`temperature_measurement_basis`	2 m air temperature from FortyGuard, NASA POWER, and Open-Meteo	Prevents comparisons between incompatible temperature measurements.
Soil	`preferred_soil_textures`	USDA SSURGO soil texture (not fetched)	Measures texture compatibility for the selected point or farm polygon.
Soil	`ph_tolerable_range`	SSURGO representative pH or farmer soil test (not fetched)	Detects unsuitable acidity or alkalinity; a soil test may override mapped evidence.
Soil	`drainage_requirement`	USDA SSURGO drainage class (not fetched)	Detects drainage mismatch and potential waterlogging risk.
Water	`water_demand`	Open-Meteo rainfall, ET0, and soil moisture + NASA precipitation (Plainview rerun pending) + SSURGO storage + irrigation	Classifies whether the location is likely to meet the crop's water demand.
Water	`drought_tolerance`	Open-Meteo rainfall/ET0/soil moisture + NASA baseline + SSURGO storage + irrigation	Adjusts suitability when the location has a likely climatic water deficit.
Water	`irrigation_requirement`	Farmer irrigation availability/reliability + calculated water deficit	Identifies crops that depend on irrigation at the selected location.
Evidence	`source_ids`	None	Links each requirement to supporting publications or authoritative guidance.
Evidence	`confidence`	Provider resolution, freshness, live/cache status, missing fields, and warnings	Contributes to recommendation confidence; it is not crop suitability itself.
Evidence	`last_reviewed`	None	Records how recently the crop profile was checked.
Evidence	`assumptions`	Farm/scenario inputs + disclosed manual Plains assignment and data gaps	Discloses variety, regional, and data assumptions used in the result.
Administration	`catalog_version`	None	Supports versioned, reproducible catalog updates.
Administration	`record_status`	None	Marks the record as draft, reviewed, or approved.

Runtime Inputs and Interpretation

Required runtime location profile

Location input	Source
Latitude and longitude or field polygon	Map selection
Texas agricultural region	Derived from coordinates
Proposed planting date	Farmer scenario input
Temperature climatology	NASA POWER
Recent and near-term heat	FortyGuard
Historical growing-season precipitation	NASA POWER
Soil texture	USDA SSURGO
Soil pH	SSURGO; optional soil-test override
Soil drainage	USDA SSURGO
Available soil water storage	USDA SSURGO
Rainfed or irrigated	Farmer
Irrigation reliability	Farmer
Groundwater context, optional	Texas Water Development Board

MVP water-risk interpretation

Climatological rainfall + soil water-storage contribution + irrigation adjustment -> approximate crop water suitability or water risk. This is not an exact irrigation-volume calculation.

Optional later refinement: add gridMET reference evapotranspiration and a documented crop-water calculation. Texas Water Development Board data may add aquifer and nearby-well context, but it cannot prove farm-level irrigation access.

Missing-data rule: if a required location input is unavailable, mark that dimension unknown and reduce confidence. Do not silently assume favorable conditions.

Authoritative Data References

FortyGuard Temperature API documentation
https://docs-api.fortyguard.com/docs

NASA POWER climatology API
https://power.larc.nasa.gov/docs/services/api/temporal/climatology/

USDA NRCS SSURGO overview
https://www.nrcs.usda.gov/resources/data-and-reports/soil-survey-geographic-database-ssurgo

USDA Soil Data Access web services
https://sdmdataaccess.sc.egov.usda.gov/webservicehelp.aspx

gridMET dataset description
https://www.climatologylab.org/gridmet.html

Texas Water Development Board data services
https://www.twdb.texas.gov/mapping/data-services.asp

Interpretation boundaries

Evidence	Boundary
NASA POWER	Regional climate context; its meteorology grid is not field truth.
SSURGO	Mapped survey estimate; a laboratory soil test can supersede representative values.
TWDB	Regional groundwater context; not proof of ownership, pumping capacity, allocation, or access.
CropSage output	Suitability and risk decision support; not a yield prediction or irrigation prescription.

Prepared 26 August 2026 from the current CropSage planning decisions.